

# Contents

|           |                                                      |           |
|-----------|------------------------------------------------------|-----------|
| <b>I</b>  | <b>Exponential Analysis</b>                          | <b>3</b>  |
| <b>1</b>  | <b>Introduction</b>                                  | <b>5</b>  |
| 1.1       | Introduction . . . . .                               | 5         |
| <b>2</b>  | <b>The Fourier Series</b>                            | <b>7</b>  |
| 2.1       | Fourier Series . . . . .                             | 7         |
| <b>3</b>  | <b>Discrete-Time Fourier Transform</b>               | <b>9</b>  |
| 3.1       | Introduction . . . . .                               | 9         |
| 3.2       | Analysis and Synthesis . . . . .                     | 9         |
| <b>4</b>  | <b>Discrete Fourier Transform</b>                    | <b>11</b> |
| 4.1       | Summary . . . . .                                    | 11        |
| <b>5</b>  | <b>Fourier Transform Properties</b>                  | <b>13</b> |
| 5.1       | Symmetry . . . . .                                   | 13        |
| 5.1.1     | Even and Odd Functions . . . . .                     | 13        |
| 5.1.2     | The Symmetry Property . . . . .                      | 13        |
| 5.1.3     | Real Functions . . . . .                             | 14        |
| 5.1.4     | Imaginary Functions . . . . .                        | 14        |
| 5.1.5     | Conjugate Symmetric Functions . . . . .              | 14        |
| 5.1.6     | Conjugate Anti-Symmetric Functions . . . . .         | 14        |
| 5.2       | Hilbert Transform Relations . . . . .                | 15        |
| 5.2.1     | Introduction . . . . .                               | 15        |
| 5.2.2     | Causal Real Functions . . . . .                      | 15        |
| 5.2.3     | Finite Causal Real Discrete-Time Functions . . . . . | 15        |
| 5.2.4     | Complex Sequences . . . . .                          | 16        |
| <b>II</b> | <b>Linear Time-Invariant Systems</b>                 | <b>17</b> |
| <b>6</b>  | <b>Linear Time-Invariant Systems</b>                 | <b>19</b> |
| 6.1       | Introduction . . . . .                               | 19        |

|            |                            |           |
|------------|----------------------------|-----------|
| <b>III</b> | <b>Random Processes</b>    | <b>21</b> |
| <b>7</b>   | <b>Random Processes</b>    | <b>23</b> |
| 7.1        | Introduction . . . . .     | 23        |
| 7.2        | Random Variables . . . . . | 23        |
| 7.3        | Random Signals . . . . .   | 25        |

Part I

# Exponential Analysis



# Chapter 1

## Introduction

### 1.1 Introduction

Complex exponentials are functions of particular interest to mathematicians, scientists, and engineers. Recall the following:

$$\frac{d}{dt}e^{st} = se^{st} \quad (1.1)$$

$$\int e^{st} dt = \frac{1}{s}e^{st} + C \quad (1.2)$$

where  $s$  is a complex constant.

The complex exponential  $e^{st}$  is an eigenfunction of differentiation with eigenvalue  $s$ . Because of this fact, complex exponentials are solutions to many differential equations and generate simple input-output relationships for systems described by differential equations.

The preceding commentary might suggest that systems described by differential equations could be simply analyzed by decomposing inputs into a sum of complex exponentials. This possibility was first explored by Joseph Fourier in 1821 as he used complex exponentials to solve problems involving heat flow. This type of analysis, called Fourier analysis in his honor, is used today in diverse fields to great effect. In this chapter, we explore the two primary ways to decompose functions into sums of complex exponentials. These are the Fourier series and transforms. In Chapter 3 we will explore which types of systems can be usefully analyzed with Fourier analysis.



## Chapter 2

# The Fourier Series

### 2.1 Fourier Series

A Fourier Series is the decomposition of a periodic function into a sum of trigonometric functions, which can be expressed equivalently in terms of sines and cosines or in terms of complex exponentials.

Below are the synthesis and analysis equations for a continuous-time function  $x(t)$  with period  $P$ .

$$x(t) = \sum_{k=-\infty}^{\infty} S[k] e^{j2\pi \frac{k}{P} t} \quad (2.1)$$

$$S[k] = \frac{1}{P} \int_0^P x(t) e^{-j2\pi \frac{k}{P} t} dt \quad (2.2)$$

Let us sample the continuous-time function  $x(t)$  with a sampling period  $T_s$ . The result is a discrete-time function  $x[n]$  such that  $x[n] = x(nT_s)$ . We can develop a Fourier series representation of  $x[n]$  from the definition of the Fourier series of  $x(t)$  shown above. Note that to guarantee the periodicity of  $x[n]$  we must constrain the sampling period  $T_s$  so that  $P = NT_s$  for some integer  $N$ .

$$\begin{aligned} S[k] &= \frac{1}{NT_s} \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{P} nT_s} \\ &= \frac{1}{NT_s} \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} \end{aligned}$$

$$\begin{aligned}
 x[n] = x(nT_s) &= \sum_{k=-\infty}^{\infty} S[k] e^{j2\pi \frac{k}{P} nT_s} \\
 &= \sum_{k=-\infty}^{\infty} S[k] e^{j2\pi \frac{k}{N} n}
 \end{aligned}$$

Below are the synthesis and analysis equations for a discrete-time function  $x[n]$  with period  $N$ .

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk\omega_0 n} \quad (2.3)$$

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n} \quad (2.4)$$



## Chapter 3

# Discrete-Time Fourier Transform

### 3.1 Introduction

### 3.2 Analysis and Synthesis

**Derivation:**

**Summary:** The analysis (DTFT) and synthesis (IDTFT) equations for the discrete-time Fourier transform are included below.

$$\text{Analysis : } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}, \quad (3.1)$$

$$\text{Synthesis : } x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega. \quad (3.2)$$



## Chapter 4

# Discrete Fourier Transform

### 4.1 Summary

Below are the analysis and synthesis equations for the discrete Fourier transform.

$$\textbf{Analysis} : X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, 0 \leq k \leq N-1, \quad (4.1)$$

$$\textbf{Synthesis} : x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, 0 \leq n \leq N-1. \quad (4.2)$$

$$W_N = e^{-j \frac{2\pi}{N}} \quad (4.3)$$



## Chapter 5

# Fourier Transform Properties

### 5.1 Symmetry

#### 5.1.1 Even and Odd Functions

To appreciate the Symmetry Property of the Fourier transform, we must understand the decomposition of functions into even and odd components. Any function  $f(t)$  has a unique decomposition into even and odd functions  $f_e$  and  $f_o$ , where

$$f_e(t) = \frac{f(t) + f(-t)}{2} \quad (5.1)$$

and

$$f_o(t) = \frac{f(t) - f(-t)}{2} \quad (5.2)$$

#### 5.1.2 The Symmetry Property

We can decompose real and imaginary parts of a complex function into even and odd sections to give four components (denoted below as RE, RO, IE, and IO). The Symmetry Property of the Fourier transform states that there is a one-to-one mapping between the four components of a complex time function and the four components of its complex frequency transform. That mapping is shown below.

$$\begin{array}{rcl}
\text{Time domain} & f & = f_{\text{RE}} + f_{\text{RO}} + i f_{\text{IE}} + i f_{\text{IO}} \\
& \updownarrow \mathcal{F} & \\
\text{Frequency domain} & F & = F_{\text{RE}} + i F_{\text{IO}} + i F_{\text{IE}} + F_{\text{RO}}
\end{array}$$

From the Symmetry Property, numerous relationships are apparent. Just a few are summarized below.

### 5.1.3 Real Functions

If  $f(t)$  is real, then its Fourier transform will have an even real component and an odd imaginary component. This type of function is called Hermitian or conjugate symmetric and satisfies the condition below.

$$F(-\omega) = F^*(\omega) \quad (5.3)$$

Conversely, if the Fourier transform of  $f(t)$  is conjugate symmetric, then  $f(t)$  is real.

### 5.1.4 Imaginary Functions

If  $f(t)$  is imaginary, then its Fourier transform will have an odd real component and an even imaginary component. This type of function is called anti-Hermitian or conjugate anti-symmetric and satisfies the condition below.

$$F(-\omega) = -F^*(\omega) \quad (5.4)$$

Conversely, if the Fourier transform of  $f(t)$  is conjugate anti-symmetric, then  $f(t)$  is imaginary.

### 5.1.5 Conjugate Symmetric Functions

If  $f(t)$  is conjugate symmetric (i.e.  $f(t) = f_{\text{RE}}(t) + f_{\text{IO}}(t)$ ), then its Fourier transform will be real. Conversely, if the Fourier transform of  $f(t)$  is real then  $f(t)$  is conjugate symmetric.

### 5.1.6 Conjugate Anti-Symmetric Functions

If  $f(t)$  is conjugate anti-symmetric (i.e.  $f(t) = f_{\text{RO}}(t) + f_{\text{IE}}(t)$ ), then its Fourier transform will be imaginary. Conversely, if the Fourier transform of  $f(t)$  is imaginary then  $f(t)$  is conjugate anti-symmetric.

## 5.2 Hilbert Transform Relations

### 5.2.1 Introduction

In system theory, Hilbert Transform Relations are a set of unique relationships between the real and imaginary parts of a complex function resulting from various constraints.

### 5.2.2 Causal Real Functions

**Derivation:** Let  $f$  be a causal, stable, real function with Fourier transform  $F(\omega)$ . Let  $f_e$  and  $f_o$  be the even and odd components of  $f$  respectively. We seek to express  $f$  in terms of  $f_e$ :

$$\begin{aligned} f_e(t) &= \frac{f(t) + f(-t)}{2} \\ 2f_e(t) &= f(t) + f(-t) \\ 2f_e(t)u(t) &= u(t)(f(t) + f(-t)) \end{aligned}$$

Because  $f$  is causal,  $f(t)$  is zero for all  $t < 0$  and  $f(-t)$  is zero for all  $t > 0$ . It follows that  $f(t)u(t) = f(t)$  and  $f(-t)u(t) = f(0)\delta(t)$ . Then

$$2f_e(t)u(t) = f(t) + \delta(t)f(0)$$

Since  $f(0) = f_e(0)$ , we can express  $f$  entirely in terms of  $f_e$ .

$$f(t) = 2u(t)f_e(t) - \delta(t)f_e(0) \quad (5.5)$$

Recall that by the Symmetry Property, the Fourier transform of  $f_e$  is real and that of  $f_o$  is imaginary. It follows that  $F$  can be determined from its real component  $F_R$ .

**Summary:** The Fourier transform of a causal, stable, real function can be uniquely determined from its real component by the following process:

1. Use the inverse Fourier transform to find  $f_e$  from  $F_R$ .
2. Recover  $f$  from  $f_e$  using (5.5).
3. Use the Fourier transform to find  $F$  from  $f$ .

It follows, of course, that  $F_I$  can be determined from  $F_R$ .

### 5.2.3 Finite Causal Real Discrete-Time Functions

Let  $f$  be a causal, finite, real, discrete-time function with DFT  $F[k]$ .

**Summary:** The discrete Fourier transform of a causal, finite, real function can be determined uniquely from its real component by the following process:

1. Compute the inverse DFT of  $F_R[k]$  to obtain the sequence

$$f_{ep}[n] = \frac{f[n] + f[(-n)_N]}{2}, 0 \leq n \leq N-1.$$

2. Compute the periodic odd part of  $f[n]$  as

$$f_{op}[n] = \begin{cases} f_{ep}[n], & 0 < n < \frac{N}{2}, \\ -f_{ep}[n], & \frac{N}{2} < n \leq N-1, \\ 0, & \text{otherwise.} \end{cases}$$

3. Compute the DFT of  $f_{op}$  to obtain  $jF_I[k]$ .

### 5.2.4 Complex Sequences

Let  $x$  be a complex sequence with a Fourier transform  $X$  such that

$$X(e^{j\omega}) = 0, -\pi \leq \omega < 0.$$

Let  $x_r$  and  $x_i$  be the real and imaginary components of  $x$  respectively such that  $x[n] = x_r[n] + ix_i[n]$ . Let  $X_r$  and  $X_i$  be the Fourier transforms of  $x_r$  and  $x_i$  respectively such that  $X(e^{j\omega}) = X_r(e^{j\omega}) + jX_i(e^{j\omega})$  (note that  $X_r$  and  $X_i$  are *not* the real and imaginary components of  $X$ ). Then

$$X(e^{j\omega}) = \begin{cases} 2X_r(e^{j\omega}), & 0 < \omega < \pi \\ 0, & -\pi \leq \omega < 0, \end{cases} \quad (5.6)$$

and

$$X(e^{j\omega}) = \begin{cases} 2jX_i(e^{j\omega}), & 0 < \omega < \pi \\ 0, & -\pi \leq \omega < 0. \end{cases} \quad (5.7)$$

Note that  $X$  cannot be recovered from either  $X_r$  or  $X_i$  alone at  $\omega = 0$ . Additionally,

$$X_i(e^{j\omega}) = H(e^{j\omega})X_r(e^{j\omega}), \quad (5.8)$$

and

$$X_r(e^{j\omega}) = -H(e^{j\omega})X_i(e^{j\omega}), \quad (5.9)$$

where

$$H(e^{j\omega}) = \begin{cases} -j, & 0 < \omega < \pi, \\ j, & \pi < \omega < 0. \end{cases} \quad (5.10)$$

A system with frequency response  $H$  is called an ideal 90-degree phase shifter, or a Hilbert Transformer.



Part II

**Linear Time-Invariant  
Systems**



## Chapter 6

# Linear Time-Invariant Systems

### 6.1 Introduction

Many systems of interest have the properties of linearity and time invariance.



**Part III**

**Random Processes**



## Chapter 7

# Random Processes

### 7.1 Introduction

### 7.2 Random Variables

In statistics, a **random variable** is a variable whose different values represent all possible outcomes of a random phenomenon. Impossible outcomes are not included in the variable's domain. Random variables may be discrete or continuous depending on whether the possible outcomes form a discrete or continuous set.

A **probability distribution function** (PDF) describes the probability associated with each value of a random variable (i.e. the probability associated with each possible outcome of the random phenomenon described by the variable). The exact meaning and use of the PDF differs for continuous and discrete random variables.

For continuous random variables, the PDF is called a **probability density function** and represents the probability density associated with each value of the random variable. For discrete random variables, the PDF is called a **probability mass function** and represents the probability mass associated with each value of the random variable.

**Cumulative probability** is the probability that the value of the random variable will fall within a given set of values. For continuous random variables, the cumulative probability is the integral of the PDF over the given set. For discrete random variables, the cumulative probability is the sum of the PDF over the given set. The cumulative probability over the entire domain of the random variable is always equal to 1.

**Example 7.1.** Suppose  $X_n$  is a random variable with a probability distribution

given by

$$p_{X_n}(x_n, n) = \begin{cases} \frac{2}{3}x_n, & 1 < x_n < 2 \\ 0, & \text{otherwise} \end{cases}$$

Find the probability  $P[X_n < 1.5]$ .

**Solution:**

$$P[X_n < 1.5] = \int_1^{1.5} \frac{2}{3}x_n dx_n = \frac{1}{3} [x_n^2]_1^{1.5} = \frac{1}{3} ((1.5)^2 - 1) = \left(\frac{1}{3}\right) \left(\frac{5}{4}\right) = \frac{5}{12}$$

**Example 7.2.** Suppose that  $X_n$  and  $X_m$  are two independent random variables with uniform probability distributions on the space given by  $0 < X_n < 1$  and  $0 < X_m < 2$ . Find  $P[X_n - X_m < 1]$ .

**Solution:**

$P[X_n - X_m < 1]$  is equal to the integral of the joint probability distribution of  $X_n$  and  $X_m$  over the set of values  $(x_n, x_m)$  satisfying  $0 < x_n < 1$ ,  $0 < x_m < 2$ , and  $x_n < 1 + x_m$ . Note that given the first two inequalities, the third inequality is always true. Hence, we will integrate the joint PDF over the entire domain of  $X_n$  and  $X_m$ .

$$P[X_n - X_m < 1] = \int_0^2 \int_0^1 p_{X_n X_m}(x_n, n, x_m, m) dx_m dx_n = 1.$$

**Expected value** is the theoretical average value of a function of a random variable over infinitely many trials of the associated random process. For continuous random variables, expected value is the integral of the function weighted by the PDF over all possible values of the random variable. For discrete random variables, expected value is the sum of the function weighted by the PDF over all possible values of the random variable. Note that expected value is a linear operator.

The **mean** of a random variable is equal to the expected value of the variable itself.

$$m_{X_c} = \mathcal{E}(X_{continuous}) = \int_{-\infty}^{\infty} x_c p_{X_c}(x_c) dx_c \quad (7.1)$$

$$m_{X_d} = \mathcal{E}(X_{discrete}) = \sum_{k=-\infty}^{\infty} x_d p_{X_d}(x_d) dx_d \quad (7.2)$$

The **variance** of a random variable is equal to the expected value of the squared difference of the variable and its mean.



$$\sigma_{X_c}^2 = \mathcal{E}((X_{continuous} - m_{X_c})^2) = \int_{-\infty}^{\infty} ((X_c - m_{X_c})^2) p_{X_c}(x_c) dx_c \quad (7.3)$$

$$\sigma_{X_d}^2 = \mathcal{E}((X_{discrete} - m_{X_d})^2) = \sum_{k=-\infty}^{\infty} ((X_d - m_{X_d})^2) p_{X_d}(x_d) dx_d \quad (7.4)$$

There are several types of independence between random variables. Two random variables are **statistically independent** if knowledge about the value of one does not given any knowledge about the value of the other. This means that the PDF of each random variable must be constant with respect to changes in the other variable's value. This condition is encapsulated in the following equation, which states that the joint probability distribution of the variables is equal to the product of the individual PDFs.

$$p_{XY}(x, y) = p_X(x)p_Y(y) \quad (7.5)$$

Two random variables are **uncorrelated** (linearly independent) if knowledge about the value of one cannot be used to linearly predict the value of the other. Some non-linear relationship may exist between the variables. Hence, lack of correlation is weaker than statistical independence. The condition for lack of correlation is defined mathematically as follows:

$$\mathcal{E}(f(X)g(Y)) = \mathcal{E}(f(X))\mathcal{E}(g(Y)). \quad (7.6)$$

One useful result of this fact is that the mean of the product of two statistically independent random variables is the product of their means.

$$m_{XY} = m_X m_Y \quad (7.7)$$

## 7.3 Random Signals

A **random signal** is a signal for which each individual sample  $x[n]$  is the outcome of a random phenomenon described by a random variable  $x_n$ . The entire signal is determined by a collection of such random variables, one for each sample time,  $-\infty < n < \infty$ . This collection of random variables is referred to as a **random process**. To completely describe a random process, we must specify the individual and joint probability distributions of each random variable.

The **autocorrelation** of a random signal measures how the signal correlates with a time-shifted version of itself. It is a function of index and time shift, calculated as the expected value of the signal at the index multiplied by the complex conjugate of the signal at a shifted index.

$$\phi_{XX}[n, m] = \mathcal{E}(X[n]X^*[n + m]) \quad (7.8)$$

The Fourier transform  $\Phi_{XX}(e^{j\omega})$  of the autocorrelation function is equal to the **power spectral density** of the signal.

A random signal is **wide-sense stationary** if its mean is time-independent and its autocorrelation function depends only on the time-shift (not the absolute time itself).

Consider a stable LTI system with real impulse response  $h[n]$ . Let  $x[n]$  be a real-valued sequence that is a sample sequence of a wide-sense stationary discrete-time process. Define  $y[n]$  as the output of the system for input  $x$ . Then

$$y[n] = \sum_{k=-\infty}^{\infty} h[n-k]x[k]$$

Let  $m_x$  and  $m_y$  be the means of  $x$  and  $y$  respectively. Then

$$m_y = m_x \sum_{k=-\infty}^{\infty} h[k], \quad (7.9)$$

or

$$m_y = H(e^{j0})m_x. \quad (7.10)$$

Let  $\phi_{XX}$  and  $\phi_{YY}$  be the autocorrelation functions for  $X$  and  $Y$  respectively. Then

$$\phi_{YY}[m] = \phi_{XX}[m] * c_{hh}[m], \quad (7.11)$$

where

$$c_{hh}[m] = \sum_{k=-\infty}^{\infty} h[k]h[m+k] = h[m] * h[-m] \quad (7.12)$$

is known as the deterministic autocorrelation function of  $h[n]$ .

Alternatively,

$$\Phi_{yy}(e^{j\omega}) = C_{hh}(e^{j\omega})\Phi_{xx}(e^{j\omega}), \quad (7.13)$$

where

$$C_{hh}(e^{j\omega}) = \|H(e^{j\omega})\|^2. \quad (7.14)$$

**Example 7.3.** Let  $x[n]$  be a sequence given by identical, independent random variables each with a uniform probability distribution from 0 to 1. Suppose  $y[n] = x[n] - x[n-1]$ .

(a) Find  $\mathcal{E}(y[n])$ .

(b) Suppose that  $\phi_{yy}[m] = \mathcal{E}(y[n+m]y^*[n])$ . Find  $\phi_{yy}[m]$ .

**Solution:**

(a)

$$\begin{aligned}\mathcal{E}(y[n]) &= \mathcal{E}(x[n] - x[n-1]) \\ &= \mathcal{E}(x[n]) - \mathcal{E}(x[n-1]) \\ &= 0.5 - 0.5 = 0.\end{aligned}$$

(b) We will seek the solution to (b) by several methods.

$$\begin{aligned}\phi_{yy}[m] &= \mathcal{E}(y[n+m]y^*[n]) \\ &= \mathcal{E}((x[n] - x[n-1])(x[n+m] - x[n+m-1])) \\ &= \mathcal{E}(x[n]x[n+m] - x[n]x[n+m-1] - x[n-1]x[n+m] + x[n-1]x[n+m-1]) \\ &= \mathcal{E}(x[n]x[n+m]) - \mathcal{E}(x[n]x[n+m-1]) - \mathcal{E}(x[n-1]x[n+m]) + \mathcal{E}(x[n-1]x[n+m-1]).\end{aligned}$$

The value of  $\phi_{yy}[m]$  varies depending on  $m$ .

1. If  $m = -1$ , then

$$\begin{aligned}\phi_{yy}[m] &= \mathcal{E}(x[n])\mathcal{E}(x[n-1]) - \mathcal{E}(x[n])\mathcal{E}(x[n-2]) - \mathcal{E}(x^2[n-1]) + \mathcal{E}(x[n-1])\mathcal{E}(x[n-2]) \\ &= \frac{1}{4} - \frac{1}{4} - \frac{1}{3} + \frac{1}{4} = -\frac{1}{12}.\end{aligned}$$

2. If  $m = 0$ , then

$$\phi_{yy}[0] = \mathcal{E}(x^2[n]) - 2\mathcal{E}(x[n])\mathcal{E}(x[n-1]) + \mathcal{E}(x^2[n-1]) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}.$$

3. If  $m = 1$ , then

$$\begin{aligned}\phi_{yy}[0] &= \mathcal{E}(x[n])\mathcal{E}(x[n+1]) - \mathcal{E}(x^2[n]) - \mathcal{E}(x[n-1])\mathcal{E}(x[n+1]) + \mathcal{E}(x[n-1])\mathcal{E}(x[n]) \\ &= \frac{1}{4} - \frac{1}{3} - \frac{1}{4} + \frac{1}{4} = -\frac{1}{12}\end{aligned}$$

4. Otherwise,

$$\begin{aligned}\phi_{yy}[m] &= \mathcal{E}(x[n])\mathcal{E}(x[n+m]) - \mathcal{E}(x[n])\mathcal{E}(x[n+m-1]) - \mathcal{E}(x[n-1])\mathcal{E}(x[n+m]) + \mathcal{E}(x[n-1])\mathcal{E}(x[n+m-1]) \\ &= \frac{1}{4} - \frac{1}{4} - \frac{1}{4} + \frac{1}{4} = 0.\end{aligned}$$

Alternatively, we can use the deterministic autocorrelation function  $c_{hh}[h]$  of  $h[n]$  to calculate  $\phi_{yy}[m]$ . Recall that  $c_{hh}[m] = h[m] * h[-m]$ . In this problem,  $h[n] = \delta[n] - \delta[n-1]$ . Then  $c_{hh}[m] = -\delta[m+1] + 2\delta[m] - \delta[m-1]$ . First, we must find  $\phi_{xx}[m]$ .

$$\begin{aligned}\phi_{xx}[m] &= \mathcal{E}(x[n]x[n+m]) \\ &= \begin{cases} \frac{1}{3}, & m = 0, \\ \frac{1}{4}, & \text{otherwise.} \end{cases}\end{aligned}$$

Then

$$\phi_{yy}[m] = \phi_{xx}[m] * c_{hh}[m] = \begin{cases} \frac{1}{6}, & m = 0, \\ -\frac{1}{12}, & m = \pm 1, \\ 0, & \text{otherwise.} \end{cases}$$

**Example 7.4.** Let  $x[n]$  be a stationary random sequence with mean  $m_x = 1$  and an autocorrelation function given by  $\phi_{xx}[m] = 2\delta[m]$ .